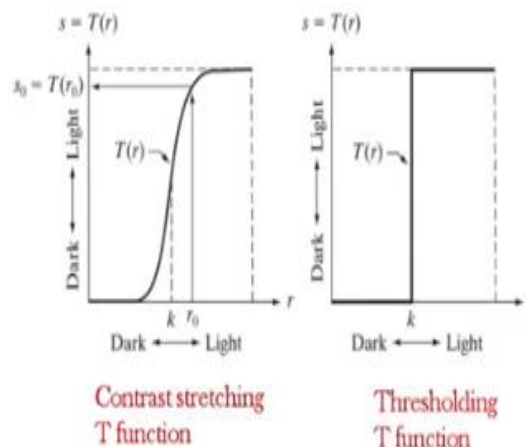
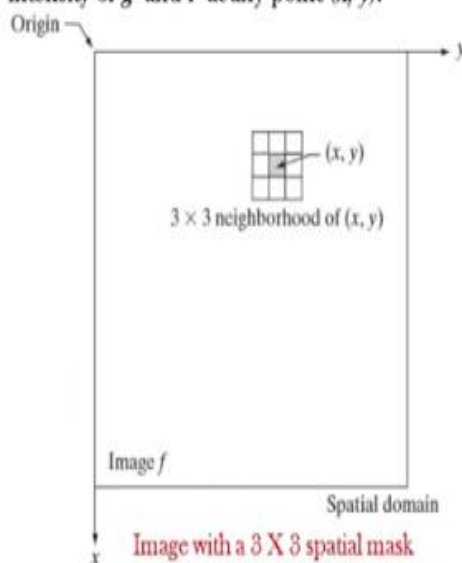


Image enhancement in spatial domain and filtering

- Image enhancement is the process of enhancing image so that the resultant image is more suitable than original for specific applications.



- The spatial domain processes can be denoted by the expression, $g(x, y) = T[f(x, y)]$, where $f(x, y)$ is the input image, $g(x, y)$ is the output image, and T is an operator on f defined over a neighborhood of point (x, y) .
- The smallest possible neighborhood is of size 1×1 . Here, g depends only on the value of f at a single point (x, y) and T becomes $s = T(r)$, where, for simplicity in notation, s and r are variables denoting, respectively, the intensity of g and f at any point (x, y) .



❖ Contrast stretching:



❖ Thresholding:



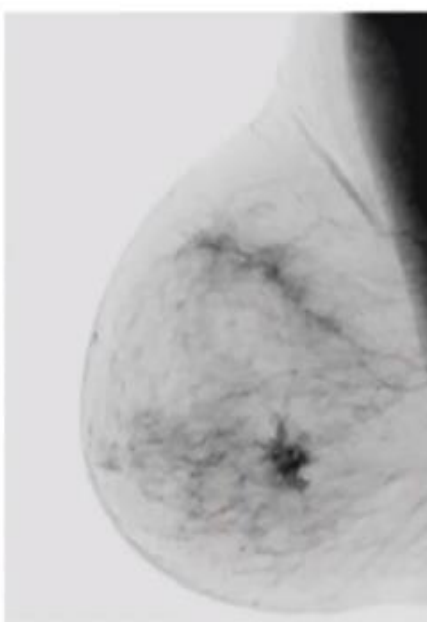
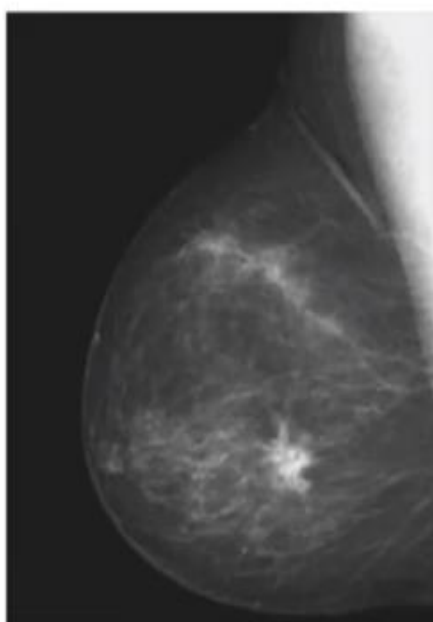
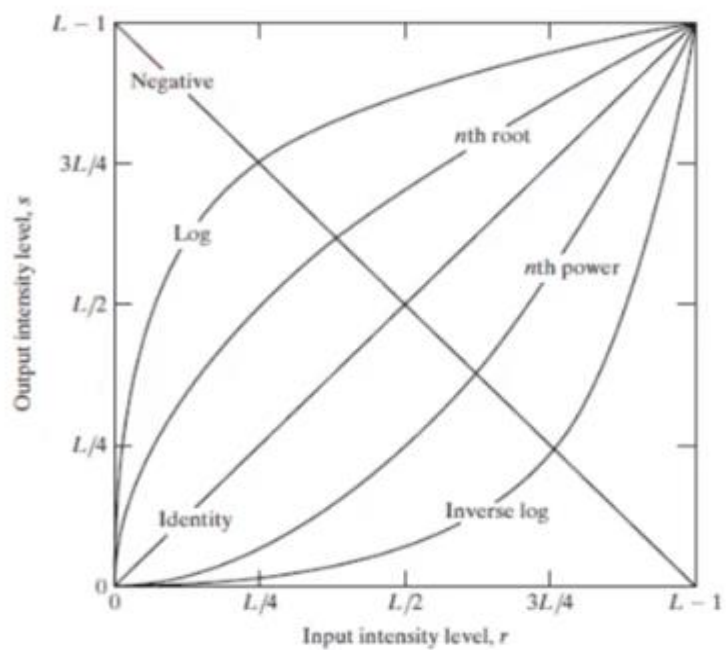
❖ Image Negatives:

- Negative image means inverting the grey levels i.e. black in the original image will become white and vice-versa.
- The transformation for image negative can be given by,

$$s = L - 1 - r$$

where, s is the output intensity value, L is the highest intensity levels and r is the input intensity value

- It is particularly suited for enhancing white or gray detail embedded in dark regions of an image, especially when the black areas are dominant in size

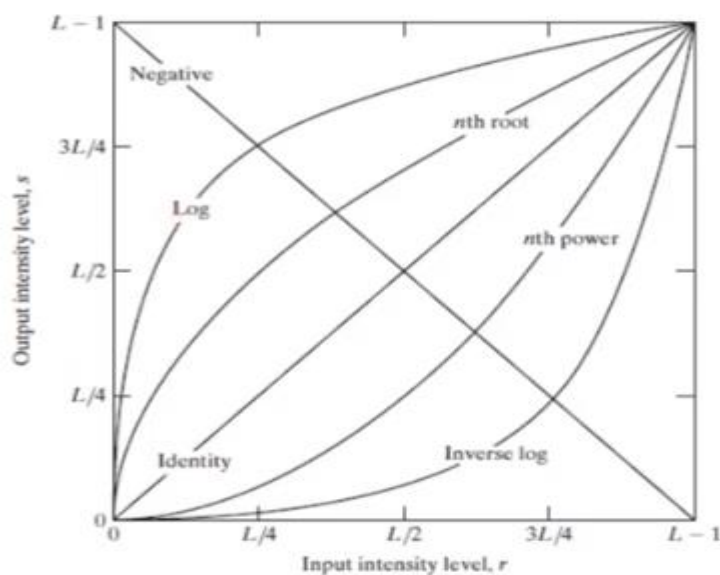


a b

FIGURE 3.4
 (a) Original digital mammogram.
 (b) Negative image obtained using the negative transformation in Eq. (3.2-1).
 (Courtesy of G.E. Medical Systems.)

❖ Log transformations:

- $s = c \log(1 + r)$
where, c is constant, s is the output intensity value and r is the input intensity value
- It maps a narrow range of low intensity values in the input into a wide range of output levels.
- The opposite is true of higher values of input levels.
- It expands the values of dark pixels in an image while compressing the higher level values.
- It compresses the dynamic range of images with large variations in pixel values.
- Log reduces contrast of brighter regions.



❖ Power-Law (Gamma) transformations:

- $s = c r^\gamma$
where c and γ are both positive constants, r =input and s = output
- With fractional values ($0 < \gamma < 1$) of gamma map a narrow range of dark input values into a wider range of output values, with the opposite being true for higher values ($\gamma > 1$) of input levels.
- $C = \text{gamma} = 1$ means it is an identity transformations.
- Variety of devices used for image capture , printing, and display respond according to a power law.
- Process used to correct these power law response phenomena is called gamma correction.
- Power-law enhances contrast of brighter regions.

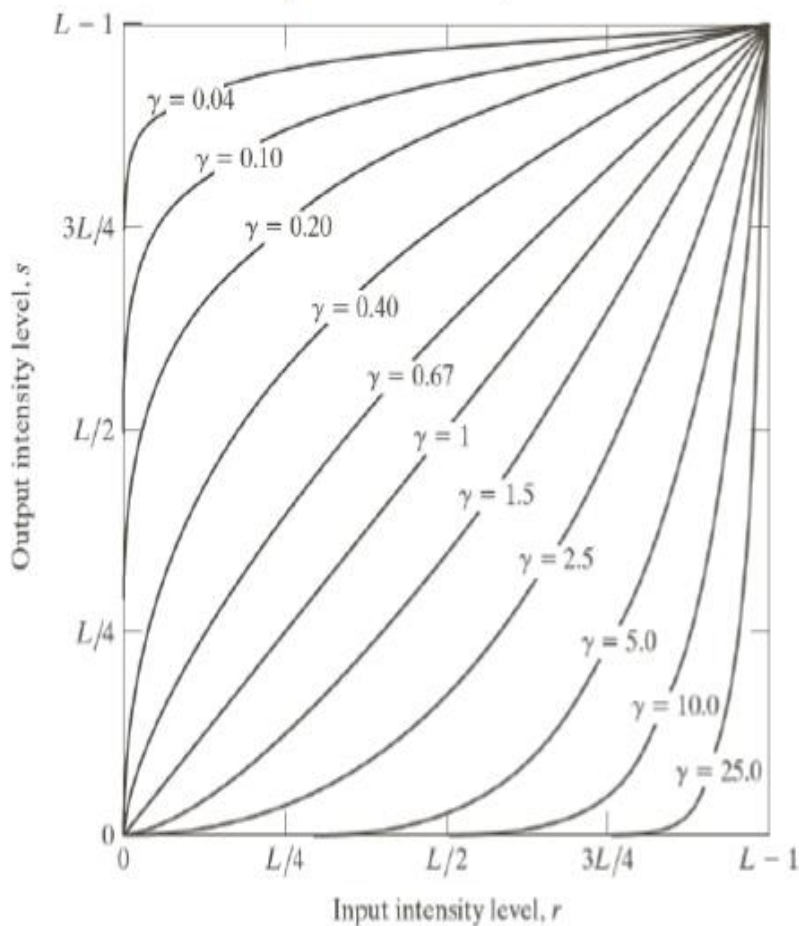
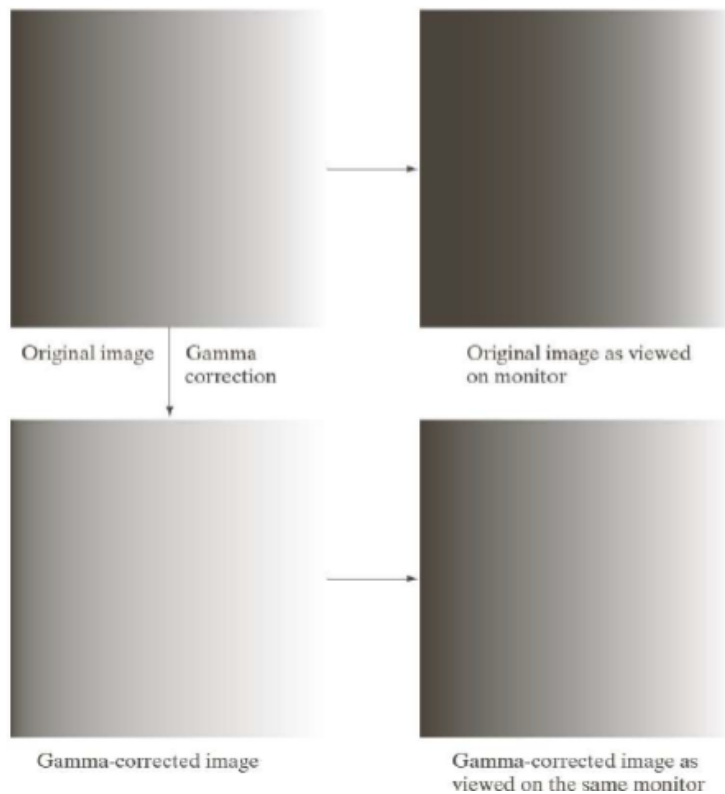


FIGURE 3.6 Plots of the equation $s = cr^\gamma$ for various values of γ ($c = 1$ in all cases). All curves were scaled to fit in the range shown.



a	b
c	d

FIGURE 3.7

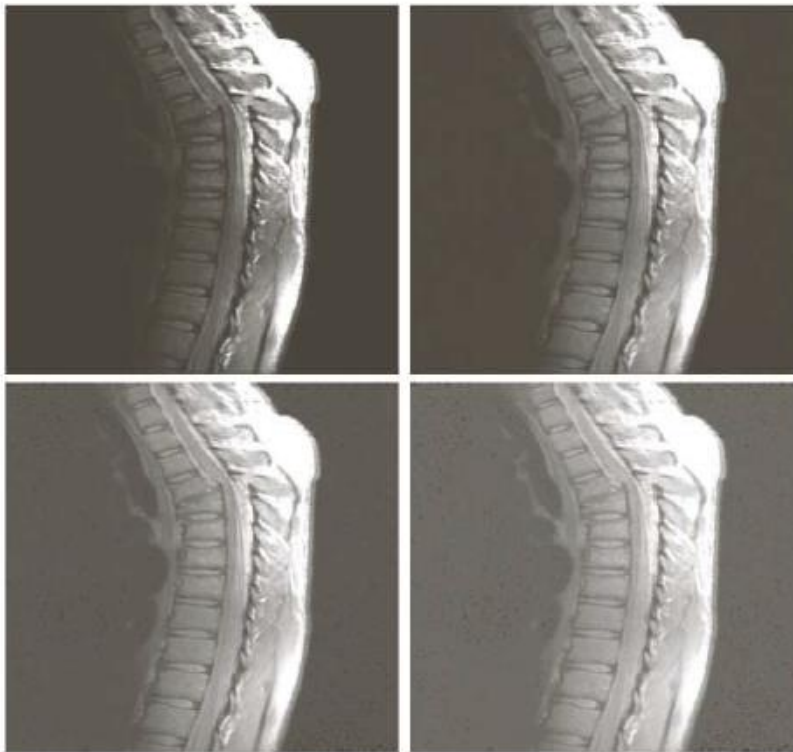
(a) Intensity ramp image. (b) Image as viewed on a simulated monitor with a gamma of 2.5. (c) Gamma-corrected image. (d) Corrected image as viewed on the same monitor. Compare (d) and (a).



a	b
c	d

FIGURE 3.9

(a) Aerial image. (b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and $\gamma = 3.0, 4.0,$ and 5.0 , respectively. (Original image for this example courtesy of NASA.)



a	b
c	d

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine.

(b)–(d) Results of applying the transformation in Eq. (3.2-3) with $c = 1$ and

$\gamma = 0.6, 0.4,$ and 0.3 , respectively.

(Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

Original Image



Gamma factor 0.6



Gamma factor 0.4



Gamma factor 0.3

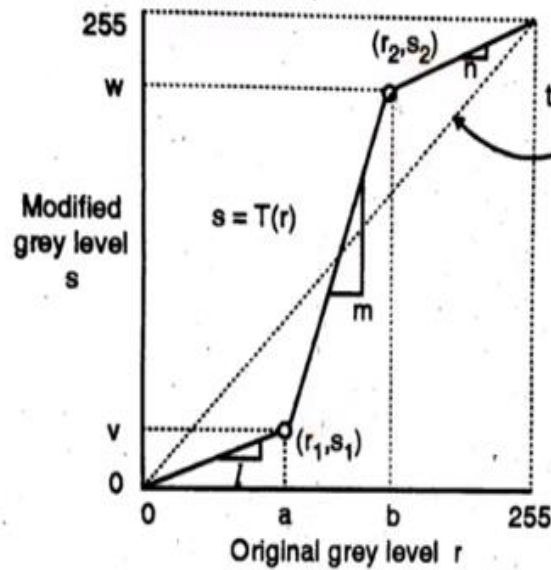


❖ Piece-wise Linear Transformation:

- Piece-wise Linear Transformation is type of gray level transformation that is used for image enhancement. It is a spatial domain method. It is used for manipulation of an image so that the result is more suitable than the original for a specific application.
- Some commonly used piece-wise linear transformations are:
 - (1) Contrast Stretching
 - (2) Intensity Level Slicing
 - (3) Bit Plane Slicing

❖ Contrast Stretching:

- Contrast is the difference between the highest grey level and low grey level of an image.
- Low contrast images can result from poor illuminations, Lack of dynamic range in the imaging sensor, or even the wrong setting of a lens aperture during image acquisition.
- It expands the range of intensity levels in an image so that it spans the full intensity range of display devices.
- The contrast can be stretched by making darker portion more darker and brighter portion more brighter.



$$s = \left(\frac{s_{max} - s_{min}}{r_{max} - r_{min}} \right) * (r - r_{min}) + s_{min}$$

For the first part,

$$s = (0.5) * (r - 0) + 0$$

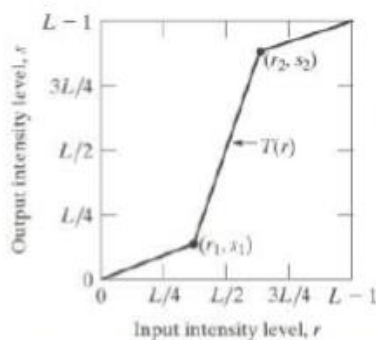
For the second part,

$$s = (2) * (r - LT) + (0.5 * LT)$$

For the third part,

$$s = (0.5) * (r - UT) + (0.5 * LT) + 2 * (UT - LT)$$

$$s = \begin{cases} l \cdot r & 0 \leq r < a \\ m \cdot (r - a) + v & a \leq r < b \\ n \cdot (r - b) + w & b \leq r < L - 1 \end{cases}$$



a b
c d

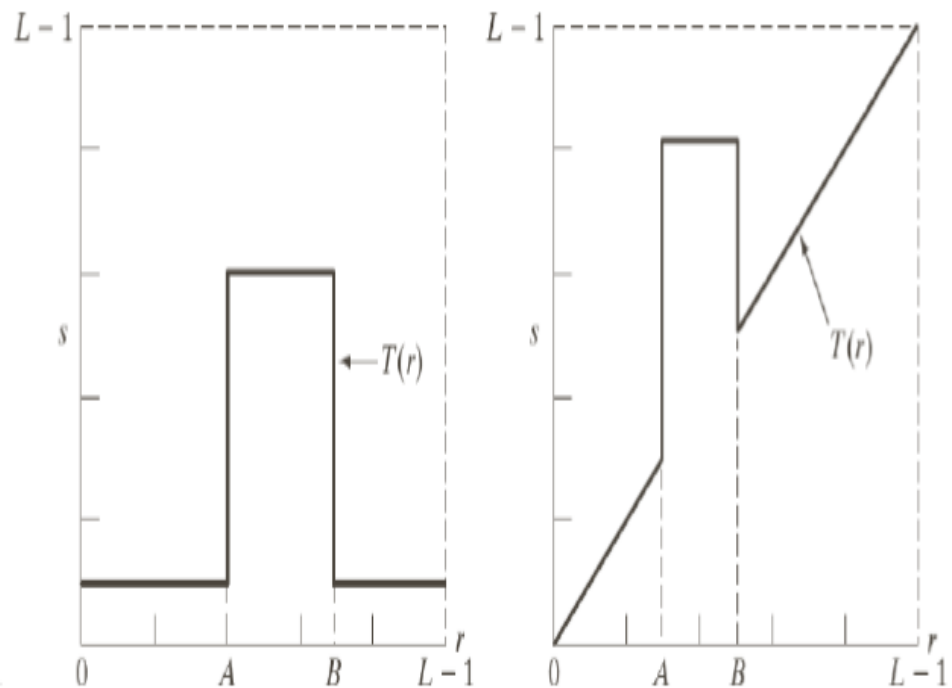
FIGURE 3.10
Contrast stretching.
(a) Form of transformation function. (b) A low-contrast image. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

❖ Intensity/Grey Level Slicing:

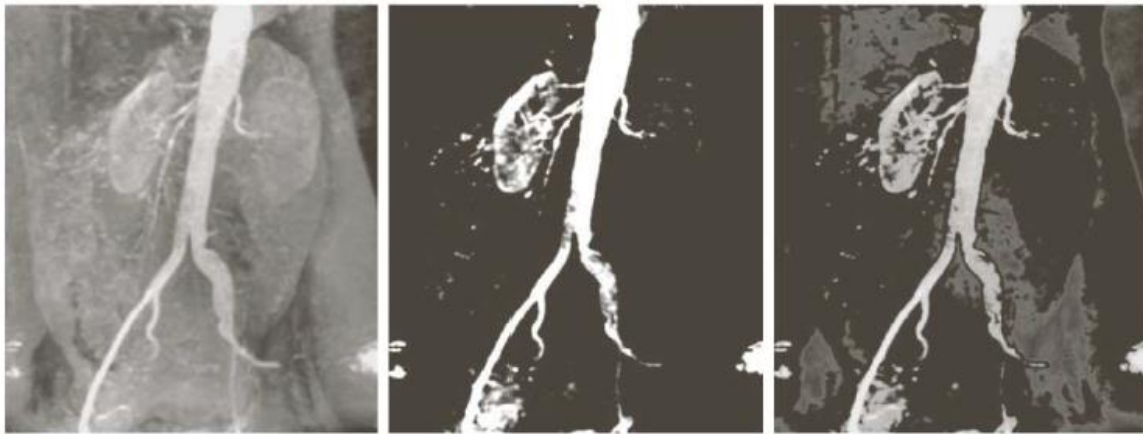
- Highlighting specific range of intensities in an image.
- Enhances features such as masses of water in satellite imagery and enhancing flaws in X-ray images.
- It can be Implemented two ways:
 - 1) To display only one value (say, white) in the range of interest and rests are black which produces binary image.
 - 2) Brightens (or darkens) the desired range of intensities but leaves all other intensity levels in the image unchanged.

a b

FIGURE 3.11 (a) This transformation highlights intensity range $[A, B]$ and reduces all other intensities to a lower level. (b) This transformation highlights range $[A, B]$ and preserves all other intensity levels.

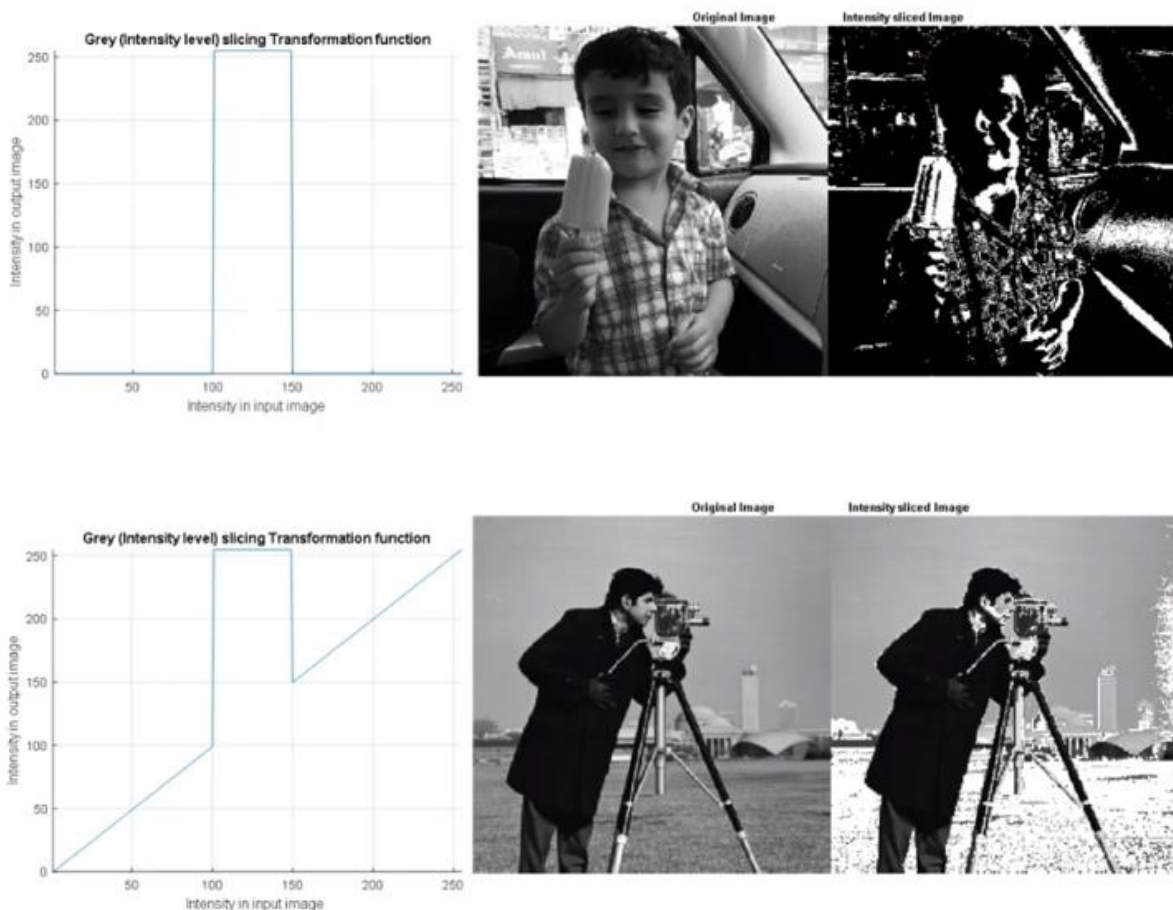


- Intensity Level Slicing



a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected area set to black, so that grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)



(3) Bit Plane Slicing:

- Pixels are digital numbers composed of bits.
- 256 gray scale image is composed of 8 bits.
- Instead of highlighting intensity level ranges, we could highlight the contribution made to total image appearance by specific bits.
- 8-bit image may be considered as being composed of eight 1-bit planes, with plane 1 containing the lowest order bit of all pixels in the image and plane 8 all the highest-order bits.

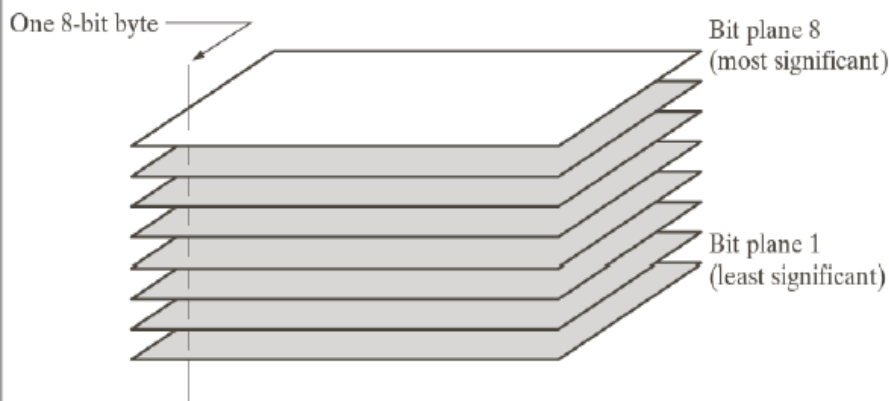


FIGURE 3.13
Bit-plane
representation of
an 8-bit image.

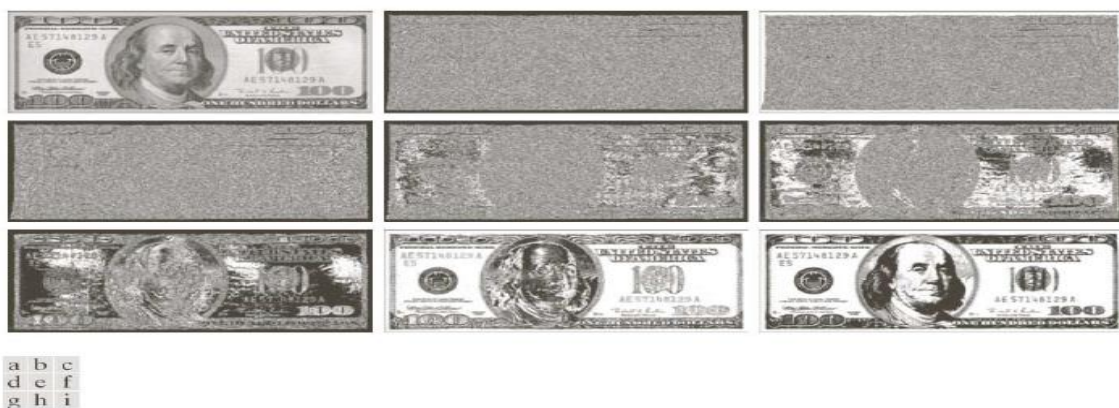


FIGURE 3.14 (a) An 8-bit gray-scale image of size 500×1192 pixels. (b) through (i) Bit planes 1 through 8, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.

Original Image



Bit plane 1



Bit plane 2



Bit plane 3



Bit plane 4



Bit plane 5



Bit plane 6



Bit plane 7



Bit plane 8

